Ruixi (Lucy) Piao

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https://ruixip.github.io/

Education and Training

University of Michigan – Ann Arbor Ann Arbor, MI
Majors: Computer Science | Data Science | Cognitive Science |
Biopsychology, Cognition, and Neuroscience (BCN) Aug 2023 – May 2027

- Honors Thesis (Advisor: Dr. David Brang): Neural integration of syntactic and visual information in sentence comprehension (completed May 2026)

The Hong Kong Polytechnic University Hong Kong
Major: Biomedical Engineering (transferred out) Aug 2022 – May 2023

Neuromatch Academy
Certificate in Computational Neuroscience Jul 2025

Conference Presentations

1. **Piao, R.,** Li, Y., Cao, C., & Brang, D. (2026, March). Neural integration of syntactic and visual temporal information in sentence comprehension. 39th Annual Conference on Human Sentence Processing (HSP), Cambridge, MA.
2. **Piao, R.,** Cao, C., Li, Y., Brissenden, J., Jahn, A., & Brang, D. (2025, September). The functional-anatomical organization of spectral and temporal restoration in auditory cortex. Seventeenth Annual Meeting of the Society for the Neurobiology of Language (SNL), Washington, DC.
3. Ma, S., **Piao, R.,** Geng, C., Hua, Y., Jones, M., & Hu, J. (2025, May). A statistical human head geometry model considering variations in head skin, skull, and brain among the whole population. The 20th Annual Injury Biomechanics Symposium (IBS), Columbus, OH.

Research Experience

University of Michigan, Multisensory Perception Lab Ann Arbor, MI
Research Assistant *(PI: Dr. David Brang)* Aug 2024 – Present
Thesis | Neural integration of syntactic and visual information in sentence comprehension

- Preprocessed raw intracranial EEG (iEEG) signals using MATLAB
- Developed a cross-validated multivariate temporal response function (mTRF) pipeline to predict iEEG responses to audio-only and audiovisual speech, testing how visual speech modulates syntactic representations
- Identified a functional dissociation in visual speech contributions: configural cues enhanced acoustic encoding under spectral degradation, while temporal cues reduced reliance on syntactic structure under temporal degradation

Other Experience

- Collected intracranial EEG and scalp EEG experimental data

University of Michigan, Transportation Research Institute Ann Arbor, MI
Research Assistant *(PI: Dr. Jingwen Hu, Dr. Monica Jones)* Jan 2024 – Dec 2025
Project | Statistical head geometry modeling from CT data across the adult population

- Co-developed a statistical head geometry model capturing anatomy variation across 181 adults to improve head injury assessment and safety equipment design
- Performed CT segmentation and landmarking to extract skin, skull, brain geometry
- Quantified shape variations using Generalized Procrustes Analysis and Principal Component Analysis, and predicted geometry from subject covariates using regression

University of Michigan, Language and Literacy Lab Ann Arbor, MI
Research Assistant *(PI: Dr. Ioulia Kovelman)* Feb 2024 – Aug 2024

- Collected fNIRS data from elementary school bilingual children (age 7-9)
- Administered phonological awareness, morphological awareness, vocabulary, and word recognition tasks in Mandarin and English

University of Michigan, Personality Processes and Outcomes Lab Ann Arbor, MI
Research Assistant *(PI: Dr. Aidan Wright)* Oct 2023 – Apr 2025

- Screened participants and conducted clinical interviews for drinking behavior studies

Research Software Development

NeuroCode: AI Copilot for EEG/fNIRS Analysis (**https://ruixip.github.io/NeuroCode/**)
Lead Developer · with Chenghao Zhou (Poeppel Lab, NYU) 2026

- Co-developed a local AI research assistant for EEG and fNIRS analysis that executes generated code in analysis kernels and grounds explanations in current data and pipeline state
- Implemented natural-language code editing and curated mappings for selected EEGLAB, Homer3, and MNE-Python functions

3D Brain Visualization Platform
Research Assistant at Brang Lab 2025

- Co-developed a 3D visualization platform that projects ECoG and sEEG electrode positions onto cortical surface meshes for interactive inspection of intracranial electrode localization using Python (nilearn) and JavaScript (Plotly.js)

Honors and Awards

- **HSP 2026 Travel Grant** ($400) Mar 2026
- **LSA Honors Grants for Research, Travel, and Special Projects** ($700) Feb 2026
- **Tanner Memorial Award** ($1500) Feb 2026
- **LSA Honors Grants for Research, Travel, and Special Projects** ($300) Nov 2025
- **Psychology Department Conference Travel Award** ($400) Oct 2025
- **Psych/BCN Undergraduate Summer Research Award** ($2500) Apr 2025
- **UMTRI Summer 2024 Student Poster Competition Third Award** Jul 2024
- **Hong Kong Jockey Club Scholarship** (HKD 800,000; full tuition and living stipend for four years; sole international recipient in 2023) Jan 2023

Internship

Michigan Medicine, Office of Development | Development Intern May – Aug 2024

Service

UMich, Department of Psychology, Student Academic Affairs Committee | Undergraduate Student Representatives Aug 2026 – Present

UMich, Department of Psychology | Peer Advisor Aug 2025 – Present

Qingyun Chinese Music Ensemble | President Aug 2025 – May 2026

UMich, Office of New Student Programs | Peer Mentor Aug 2024 – May 2026

UMich, Program on Intergroup Relations | Facilitator Jan 2024 – Aug 2025

Horses Offering People Enrichment | Equine Assisted Therapy Volunteer Jul 2023

Technical Skills

Programming Languages: C/C++, Python, MATLAB, R, HTML/CSS, JavaScript, Java, SQL

Neural Data Collection: fNIRS, scalp and intracranial EEG

Neural Data Analysis: EEGLAB, MNE, Freesurfer, SPM, CONN, nilearn

Behavioral Experiment: Structured Clinical Interview for DSM-5, REDCap, behavioral testing on adults and children (age 7-9)

Hardware: Arduino

Others: Git, Unix, Shell, CAD, GraphPad Prism, HyperMesh, Mimics, Praat, LaTeX, HPC

Languages: English (proficient), Mandarin (native), Cantonese (beginner), Korean (beginner)